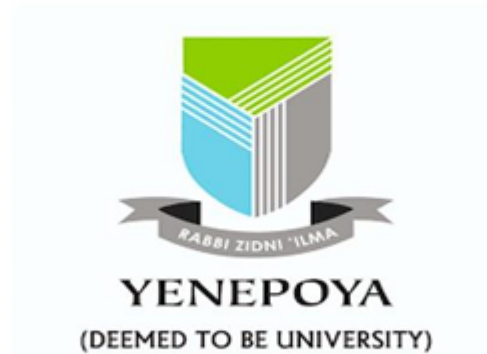




PROJECT REPORT

Fake News Detection Using Social Media Text
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PROJECT DOMAIN: ARTIFICIAL INTELLIGENCE / NLP

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1. Introduction

The rapid spread of misinformation on social media platforms has made fake news detection a critical challenge. This project focuses on developing a Fake News Detection System using Natural Language Processing (NLP) techniques to classify social media text as real or fake. The system aims to assist users in identifying misleading information and improving awareness of content authenticity.

2. Problem Statement

Social media platforms enable fast dissemination of information without verification, to the spread of fake or misleading news. Manual verification is time-consuming and inefficient. There is a need for an automated system that can analyze textual content and classify it as real or fake with reasonable accuracy.

3. Objective of the Project

- To analyze social media text using NLP techniques
- To classify news content as real or fake
- To provide a simple and interactive user interface
- To handle invalid or empty inputs effectively

4. System Architecture

- **User Interface:** Streamlit-based frontend for text input
- **Preprocessing Module:** Cleans and prepares input text
- **Classification Model:** NLP-based model for prediction
- **Output Module:** Displays classification result to the user

5. Methodology

1. Collection of social media text data
2. Preprocessing of text data (cleaning, tokenization)
3. Training of an NLP-based classification model
4. Integration of the trained model with a Streamlit interface
5. Testing and evaluation of system behavior

6. Tools and Technologies Used

- **Programming Language:** Python
- **Libraries:** NLP and machine learning libraries
- **Frontend Framework:** Streamlit
- **Development Environment:** Local system

7. Implementation

The system accepts text input through a web interface. The input is processed and passed to the trained NLP model, which predicts whether the news is real or fake. The result is displayed instantly to the user. Error handling ensures proper response for empty or invalid inputs.

8. Results and Discussion

The system successfully classifies real and fake news inputs during testing. It demonstrates stable behavior across multiple test cases and handles invalid input scenarios effectively. The results indicate that the approach is suitable for basic fake news detection tasks.

9. Limitations

- Ambiguous or sarcastic content may be misclassified
- Accuracy depends on training data quality
- Performance may vary for long or complex text inputs

10. Future Enhancements

- Training on larger and more diverse datasets
- Supporting multiple languages
- Improving accuracy using advanced transformer models
- Deploying the system on a cloud platform

11. Conclusion

The Fake News Detection System demonstrates the effective use of NLP techniques for classifying social media text. The project meets its objectives and provides a foundation for further research and enhancement in automated misinformation detection.